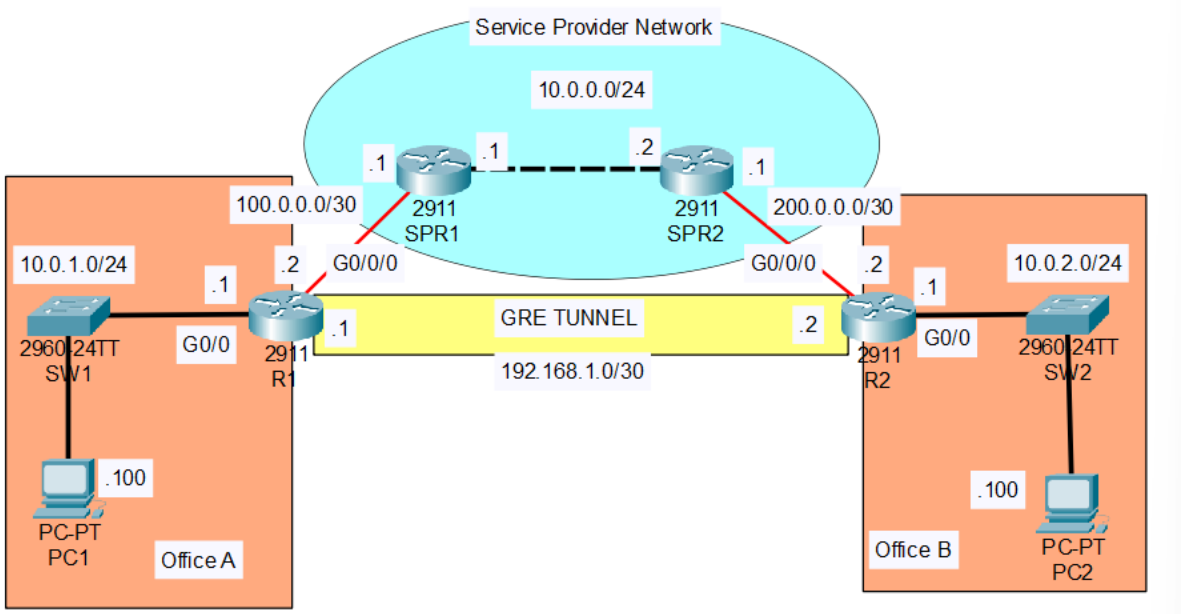
Network Topology:



Instructions and actions:

1. Configure a GRE tunnel to connect R1 and R2.
2. Configure OSPF on the tunnel interfaces of R1 and R2 to allow PC1 and PC2 to communicate.

   It is literally fun to search and configure everything on your own without any cheat sheet of commands.

   R1 CLI:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#default-information originate
R1(config-router)#ip route 0.0.0.0 0.0.0.0 100.0.0.1
R1(config)#router ospf 1
R1(config-router)#network 100.0.0.1 0.0.0.3 area 1
R1(config-router)#network 10.0.1.0 0.0.0.255 area 1
R1(config-router)#
```

   SPR1 CLI:

```
SPR1>en
Password:
SPR1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SPR1(config)#router ospf 1
SPR1(config-router)#network 100.0.0.0 0.0.0.3 area 1
SPR1(config-router)#
00:20:30: %OSPF-5-ADJCHG: Process 1, Nbr 100.0.0.2 on
GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
```

R2 CLI:

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no ip route 0.0.0.0 0.0.0.0 200.0.0.1
R2(config)#router ospf 1
R2(config-router)#default-information originate
R2(config-router)#ip route 0.0.0.0 0.0.0.0 200.0.0.1
R2(config)#router ospf 1
R2(config-router)#network 200.0.0.0 0.0.0.3 area 0
R2(config-router)#
00:11:13: %OSPF-5-ADJCHG: Process 1, Nbr 200.0.0.1 on
GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

R2(config-router)#network 10.0.2.0 0.0.0.255 area 0
R2(config-router)#
```

SPR2 CLI:

```
SPR2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SPR2(config)#router ospf 1
SPR2(config-router)#network 200.0.0.0 0.0.0.3 area 0
SPR2(config-router)#
00:11:13: %OSPF-5-ADJCHG: Process 1, Nbr 200.0.0.2 on
GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
```

PC1 can ping 100.0.0.1, and PC2 can ping 200.0.0.1. Therefore, individual networks are established.

Also, R1 can ping 200.0.0.2 (public IP of R2) and R2 can ping the public IP of R1 (100.0.0.2)

But PC1 can't ping PC2. So here we have the solution – GRE tunnels.

R1 CLI:

```
R1>en
R1#ping ?
  WORD  Ping destination address or hostname
  ip    IP echo
  ipv6  IPv6 echo
  <cr>
R1#ping 200.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.0.0.2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
0/2/11 ms

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int tunnel 1

R1(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up

R1(config-if)#ip address 192.168.1.1 255.255.255.252
R1(config-if)#tunnel source g0/0/0
R1(config-if)#tunnel destination 200.0.0.2 ?
  <cr>
R1(config-if)#tunnel destination 200.0.0.2
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to up

R1(config-if)#exit
R1(config)#ip route 10.0.2.0 255.255.255.0 192.168.1.2
R1(config)#
```

R2 CLI:

```
R2>en
R2#ping 100.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.0.0.2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3
ms

R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int tunnel 1

R2(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up

R2(config-if)#ip address 192.168.1.2 255.255.255.252
R2(config-if)#tunnel source g0/0/0
R2(config-if)#tunnel destination 100.0.0.2
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to up

R2(config-if)#exit
R2(config)#ip route 10.0.1.0 255.255.255.0 192.168.1.1
R2(config)#
```

PC1 Command Prompt:

```
C:\>ping 10.0.2.100

Pinging 10.0.2.100 with 32 bytes of data:

Reply from 10.0.2.100: bytes=32 time<1ms TTL=126
Reply from 10.0.2.100: bytes=32 time<1ms TTL=126
Reply from 10.0.2.100: bytes=32 time<1ms TTL=126
Reply from 10.0.2.100: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.2.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2 Command Prompt:

```
C:\>ping 10.0.1.100

Pinging 10.0.1.100 with 32 bytes of data:

Reply from 10.0.1.100: bytes=32 time<1ms TTL=126
Reply from 10.0.1.100: bytes=32 time<1ms TTL=126
Reply from 10.0.1.100: bytes=32 time=10ms TTL=126
Reply from 10.0.1.100: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Basically, GRE tunnels use the same path, but they are created virtually so that traffic is encapsulated in a GRE header over the entire route.